



Inverse Trigonometric Functions : RACE-01

Single Correct Question

1. If α, β are roots of the equation $6x^2 + 11x + 3 = 0$, then

- (A) both $\cos^{-1} \alpha$ and $\cos^{-1} \beta$ exists
 (B) both $\operatorname{cosec}^{-1} \alpha$ and $\operatorname{cosec}^{-1} \beta$ exists
 (C) both $\cot^{-1} \alpha$ and $\cot^{-1} \beta$ exists
 (D) none of these

MIT154

2. $\sin^{-1}\left(\sin \frac{33\pi}{7}\right) + \cos^{-1}\left(\cos \frac{46\pi}{7}\right) + \tan^{-1}\left(-\tan \frac{13\pi}{8}\right) + \cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)$ is equal to

- (A) $\frac{11\pi}{7}$ (B) $\frac{9\pi}{7}$ (C) $\frac{13\pi}{7}$ (D) $\frac{17\pi}{7}$

MIT155

3. A function $g(x)$ satisfies the following conditions

- (i) Domain of g is $(-\infty, \infty)$ (ii) Range of g is $[-1, 7]$
 (iii) g has a period π and (iv) $g(2) = 3$

Then which of the following may be possible.

- (A) $g(x) = 3 + 4 \sin(n\pi + 2x - 4), n \in \mathbb{I}$ (B) $g(x) = 3 - 8 \sin(n\pi + 2x - 4), n \in \mathbb{I}$
 (C) $g(x) = 3 + 4 \cos(n\pi + 2x - 4), n \in \mathbb{I}$ (D) $g(x) = 3 - 8 \cos(n\pi + 2x - 4), n \in \mathbb{I}$

MIT156

4. Find the value of $\sin^{-1}(\cos(\sin^{-1} x)) + \cos^{-1}(\sin(\cos^{-1} x))$

- (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) π (D) 2π

MIT157

Matrix Match Type Question

5. Find the domain of each of the following functions :

Column- I

Column- II

(P) $f(x) = \sqrt{1-2x} + 3\sin^{-1}\left(\frac{3x-1}{2}\right)$

(1) ϕ

(Q) $f(x) = 2^{\sin^{-1} x} + \frac{1}{\sqrt{x-2}}$

(2) $\left[-\frac{1}{3}, \frac{1}{2}\right]$

(R) $f(x) = \arccos \frac{2x}{1+x}$

(3) $\{1, -1\}$

(S) $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$

(4) $-1/3 \leq x \leq 1$

MIT158

Inverse Trigonometric Functions

6. Find the domain of definition of the following functions.

where $\{ \cdot \}$ represents fractional part function

Column- I	Column- II
(P) $f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) - \log_{10}(4-x)$	(1) $[3/2, 2]$
(Q) $f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$	(2) $(-2, 2) - \{-1, 0, 1\}$
(R) $f(x) = \sqrt{3-x} + \cos^{-1}\left(\frac{3-2x}{5}\right) + \log_6(2 x -3) + \sin^{-1}(\log_2 x)$	(3) $1 \leq x < 4$
(S) $f(x) = e^{\sin^{-1}\left(\frac{x}{2}\right)} + \tan^{-1}\left[\frac{x}{2}-1\right] + \ln(\sqrt{x-[x]})$	(4) $x \in \left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$

MIT159

7. Find the range of each of the following functions:

Column- I	Column- II
(P) $f(x) = \ln(\sin^{-1} x)$	(1) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$
(Q) $f(x) = \sin^{-1}\left(\frac{\sqrt{3x^2+1}}{5x^2+1}\right)$	(2) $[0, \pi]$
(R) $f(x) = \cos^{-1}\left(\frac{(x-1)(x+5)}{x(x-2)(x-3)}\right)$	(3) $[0, \pi/2]$
(S) $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$	(4) $(-\infty, \ell n \pi/2]$

MIT160

Integer Type Question

8. If $\sin^{-1} x + \sin^{-1} y = \pi$ and $x = ky$, then find the value of $(39)^{2k} + 5^k$.

MIT161

ANSWER KEY

Question	1	2	3	4	5	6
Answer	C	C	A	A	P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 4; S $\rightarrow$ 3	P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 1; S $\rightarrow$ 2
Question	7				8	
Answer	P $\rightarrow$ 4; Q $\rightarrow$ 3; R $\rightarrow$ 2; S $\rightarrow$ 1				1526	



Inverse Trigonometric Functions : RACE-02

Single Correct Question

1. If $\sin^{-1} x + \cot^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{2}$, then x is equal to
 (A) 0 (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{\sqrt{3}}{2}$ MIT162
2. The value of $\cot \left(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right)$ is equal to-
 (A) $\frac{6}{17}$ (B) $\frac{3}{17}$ (C) $\frac{4}{17}$ (D) $\frac{5}{17}$ MIT163
3. If $\sin^{-1} \left(\frac{x}{5} \right) + \operatorname{cosec}^{-1} \left(\frac{5}{4} \right) = \frac{\pi}{2}$, then a value of x is-
 (A) 1 (B) 3 (C) 4 (D) 5 MIT164
4. If $\pi \leq x \leq 2\pi$, then $\cos^{-1}(\cos x)$ is equal to
 (A) x (B) $\pi - x$ (C) $2\pi + x$ (D) $2\pi - x$ MIT165
5. If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, then $4x^2 - 4xy \cos \alpha + y^2$ is equal to -
 (A) $2\sin 2\alpha$ (B) 4 (C) $4 \sin^2 \alpha$ (D) $-4 \sin^2 \alpha$ MIT166

Multiple Correct Question

6. For what values of α , the equation, $(\sin^{-1} x)^3 + (\cos^{-1} x)^3 = \alpha\pi^3$ has no roots.
 (A) $\left(-\infty, \frac{1}{32} \right)$ (B) $\left(\frac{7}{8}, \infty \right)$ (C) $\mathbb{R}$ (D) $\mathbb{R} - \{1\}$ MIT167

Matrix Match Type Question

7. Match the column:

Column -I	Column -II
(P) $\tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$	(1) $\frac{\sqrt{2}}{4}$
(Q) $\cos \left(\sin^{-1} \frac{4}{5} + \cos^{-1} \frac{2}{3} \right)$	(2) $2\sqrt{2}$
(R) $\sin \left(\frac{1}{4} \arcsin \frac{\sqrt{63}}{8} \right)$	(3) $\frac{17}{6}$
(S) $\cot \left[\cos^{-1} \left(\frac{3}{4} \right) + \sin^{-1} \left(\frac{3}{4} \right) - \sec^{-1} 3 \right]$	(4) $\frac{6-4\sqrt{5}}{15}$

MIT168

Inverse Trigonometric Functions

8. Find the value of the following inverse trigonometric expressions :

Column - I		Column - II	
(P)	$\cos^{-1}(\cos 10)$	(1)	$2\pi - 6$
(Q)	$\tan^{-1}(\tan(-6))$	(2)	$4\pi - 10$
(R)	$\cot^{-1}(\cot(-8))$	(3)	$\frac{17\pi}{20}$
(S)	$\cos^{-1}\left(\frac{1}{\sqrt{2}}\left(\cos\frac{9\pi}{10} - \sin\frac{9\pi}{10}\right)\right)$	(4)	$3\pi - 8$

MIT169

Subjective Type Question

9. Find the set of values of 'a' for which the equation $2 \cos^{-1} x = a + a^2(\cos^{-1} x)^{-1}$ posses a solution.

MIT170

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	B	A	B	D	C	A,B	$P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
Question	8				9		
Answer	$P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$				$a \in [-2\pi, \pi] - \{0\}$		



Inverse Trigonometric Functions : RACE-03

Single Correct Question

1. Find all values of k for which there is a triangle whose angles have measure $\tan^{-1}\left(\frac{1}{2}\right), \tan^{-1}\left(\frac{1}{2}+k\right)$ and $\tan^{-1}\left(\frac{1}{2}+2k\right)$.

(A) 2 (B) 3 (C) $\frac{11}{4}$ (D) 5

MIT171

2. If x, y, z are in A.P. and $\tan^{-1} x, \tan^{-1} y$ and $\tan^{-1} z$ are also in A.P., then
(A) $x = y = z$ (B) $2x = 3y = 6z$ (C) $6x = 3y = 2z$ (D) $6x = 4y = 3z$

MIT172

3. Find the simplest value of $f(x) = \arccos x + \arccos\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$, $x \in \left(\frac{1}{2}, 1\right)$

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{9}$ (D) π

MIT173

4. If $x \geq 0$ and $\theta = \sin^{-1} x + \cos^{-1} x - \tan^{-1} x$, then

(A) $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{4}$ (B) $0 \leq \theta \leq \frac{\pi}{4}$ (C) $0 \leq \theta < \frac{\pi}{2}$ (D) $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$

MIT174

5. If α is a real root of the equation $x^3 + 3x - \tan 2 = 0$, then $\cot^{-1} \alpha + \cot^{-1} \frac{1}{\alpha} - \frac{\pi}{2}$ can be equal to

(A) 0 (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{2}$

MIT175

6. Let $f : (-1, 1) \rightarrow B$, be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$, then f is both one-one and onto when B is the interval-

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) $\left(0, \frac{\pi}{2}\right)$ (D) $\left[0, \frac{\pi}{2}\right)$

MIT176

7. Let $\tan^{-1} y = \tan^{-1} x + \tan^{-1} \left(\frac{2x}{1-x^2}\right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then a value of y is :

(A) $\frac{3x-x^3}{1+3x^2}$ (B) $\frac{3x+x^3}{1+3x^2}$ (C) $\frac{3x-x^3}{1-3x^2}$ (D) $\frac{3x+x^3}{1-3x^2}$

MIT177

Inverse Trigonometric Functions

Multiple Correct Type Question

8. Find the value of $\sin\left(\tan^{-1} a + \tan^{-1} \frac{1}{a}\right); a \neq 0$

- (A) 1 if $a > 0$ (B) 0 (C) -1 if $a < 0$ (D) 1

MIT178

9. Which of the following pair(s) are identical -

- (A) $y = \tan(\cos^{-1} x); y = \frac{\sqrt{1-x^2}}{x}$ (B) $y = \tan(\cot^{-1} x); y = \frac{1}{x}$
 (C) $y = \sin(\arctan x); y = \frac{x}{\sqrt{1+x^2}}$ (D) $y = \cos(\arctan x); y = \sin(\operatorname{arccot} x)$

MIT179

Matrix Match Type Question

10. Match the column :

Column-I		Column-II	
(P)	$\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right)$	(1)	$\tan^{-1}\left(\frac{1}{2}\right)$
(Q)	$\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right)$	(2)	$\frac{\pi}{4}$
(R)	$2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25}$	(3)	$\frac{\pi}{6}$
(S)	$\arccos\sqrt{\frac{2}{3}} - \arccos\frac{\sqrt{6}+1}{2\sqrt{3}}$	(4)	π

MIT180

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	C	A	B	D	C	B	C	A,C	A,B,C,D
Question	10								
Answer	$P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$								



Inverse Trigonometric Functions : RACE-04

Single Correct Question

1. The numerical value of $\cot\left(2\sin^{-1}\frac{3}{5} + \cos^{-1}\frac{3}{5}\right)$ is

- (A) $\frac{-4}{3}$ (B) $\frac{-3}{4}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

MIT181

2. Simplify $\tan\left\{\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right\}$, if $x > y > 1$.

- (A) $\frac{1-xy}{x-y}$ (B) $\frac{1+xy}{x-y}$ (C) $\frac{1+xy}{x+y}$ (D) $\frac{1-xy}{x+y}$

MIT182

3. Solve $\tan^{-1}\frac{x-1}{x+1} + \tan^{-1}\frac{2x-1}{2x+1} = \tan^{-1}\frac{23}{36}$

- (A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{7}{3}$ (D) $\frac{3}{7}$

MIT183

4. Solve $2\tan^{-1}x = \cos^{-1}\frac{1-a^2}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2}$ ($a > 0, b > 0$)

- (A) $x = \frac{a-b}{1+ab}$ (B) $x = \frac{a+b}{1-ab}$ (C) $x = \frac{a+b}{1+ab}$ (D) $x = \frac{a-b}{1-ab}$

MIT184

Multiple Correct Type Question

5. $\tan^{-1}x, x < 0$ is equal to

- (A) $-\pi + \cot^{-1}\frac{1}{x}$ (B) $\sin^{-1}\frac{x}{\sqrt{1+x^2}}$ (C) $-\cos^{-1}\frac{1}{\sqrt{1+x^2}}$ (D) $\cos^{-1}\frac{1}{\sqrt{1+x^2}}$

MIT185

6. $\cos^{-1}x, -1 < x < 0$ is equal to

- (A) $\sec^{-1}\frac{1}{x}$ (B) $\pi - \sin^{-1}\sqrt{1-x^2}$ (C) $\pi + \tan^{-1}\frac{\sqrt{1-x^2}}{x}$ (D) $\cot^{-1}\frac{x}{\sqrt{1-x^2}}$

MIT186

Inverse Trigonometric Functions

Matrix Match Type Question

7. Prove that

Column - I		Column - II	
(P)	$\sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{8}{17} \right)$	(1)	$\frac{\pi}{2}$
(Q)	$\tan^{-1} \frac{3}{4} + \sin^{-1} \frac{5}{13}$	(2)	$\frac{\pi}{4}$
(R)	$2 \left[\sin^{-1} \left(\frac{1}{\sqrt{5}} \right) + \cot^{-1} 3 \right]$	(3)	$\sin^{-1} \frac{77}{85}$
(S)	$\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{5} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{8} \right)$	(4)	$\cos^{-1} \frac{33}{65}$

MIT187

Integer Type Question

8. If $\tan^{-1} x + \cot^{-1} \frac{1}{y} + 2\tan^{-1} z = \pi$, then $x + y + \mu z = xz^2 + yz^2 + \lambda xyz$. Find $\lambda + \mu$

MIT188

9. If $a > 0, b > 0, \tan^{-1} \left(\frac{a}{x} \right) + \tan^{-1} \left(\frac{b}{x} \right) = \frac{\pi}{2}$ then $x = (ab)^{\frac{1}{n}}$ find $n (n \in \mathbb{N})$

MIT189

Subjective Type Question

10. Prove the following results :

- (i) $2 \tan^{-1} \left(\frac{1}{5} \right) + \tan^{-1} \left(\frac{1}{8} \right) = \tan^{-1} \left(\frac{4}{7} \right)$
- (ii) $2 \sin^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{17}{31} \right) = \frac{\pi}{4}$

MIT190

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	B	B	B	A	A,B,C	A,B,C,D	$P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
Question	8	9					
Answer	4	2					



Inverse Trigonometric Functions : RACE-05

Single Correct Question

1. Solve the inequality $\tan^{-1} x > \cot^{-1} x$.

(A) $(1, \infty)$ (B) $(-2, 0)$
(C) $(-\infty, 1)$ (D) None of these

MIT191

2. Complete solution set of inequation $(\cos^{-1} x)^2 - (\sin^{-1} x)^2 > 0$, is

(A) $\left[0, \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1, \frac{1}{\sqrt{2}}\right)$
(C) $(-1, \sqrt{2})$ (D) none of these

MIT192

3. All x satisfying the inequality $(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$, lie in the interval :-

(A) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$ (B) $(\cot 5, \cot 4)$
(C) $(\cot 2, \infty)$ (D) $(-\infty, \cot 5) \cup (\cot 2, \infty)$

MIT193

4. Solve for x : $\sin^{-1} \left(\sin \left(\frac{2x^2 + 4}{1 + x^2} \right) \right) < \pi - 3$.

(A) $x \in (-2, -1)$ (B) $x \in (-1, 1)$ (C) $x \in (2, 4)$ (D) $x \in (-3, -1)$

MIT194

5. The solution of the equation $\sin^{-1} \left(\tan \frac{\pi}{4} \right) - \sin^{-1} \left(\sqrt{\frac{3}{x}} \right) - \frac{\pi}{6} = 0$ is

(A) $x = 2$ (B) $x = -4$ (C) $x = 4$ (D) $x = 3$

MIT195

6. Number of solutions of the equation $\cot^{-1} \sqrt{4 - x^2} + \cos^{-1}(x^2 - 5) = \frac{3\pi}{2}$ is :

(A) 2 (B) 4 (C) 6 (D) 8

MIT196

Multiple Correct Type Question

7. Find all the positive integral solutions (x, y) of, $\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1 + y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$.

(A) (1, 2) (B) (2, 3) (C) (2, 7) (D) (1, 4)

MIT197

Inverse Trigonometric Functions

Matrix Match Type Question

8. Solve for x :

Column - I

(P) $\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}$

(Q) $\tan^{-1} \frac{1}{1+2x} + \tan^{-1} \frac{1}{1+4x} = \tan^{-1} \frac{2}{x^2}$

(R) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

(S) $3\cos^{-1} x = \sin^{-1}(\sqrt{1-x^2}(4x^2-1))$

Column - II

(1) $x = 3$

(2) $x = \frac{1}{2}\sqrt{\frac{3}{7}}$

(3) $\left[\frac{\sqrt{3}}{2}, 1\right]$

(4) $x = 0, \frac{1}{2}, -\frac{1}{2}$

MIT198

Subjective Type Question

9. Solve the following inequalities:

(i) $\arccot^2 x - 5 \arccot x + 6 > 0$

(ii) $\arcsin x > \arccos x$

(iii) $\tan^2(\arcsin x) > 1$

MIT199

10. Solve the following system of inequalities :

$4 \arctan^2 x - 8 \arctan x + 3 < 0$ & $4 \arccot x - \arccot^2 x - 3 \geq 0$

MIT200

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	B	C	B	C	A	A,C	$P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

9. (i) $(\cot 2, \infty) \cup (-\infty, \cot 3)$ (ii) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (iii) $\left(\frac{\sqrt{2}}{2}, 1\right) \cup \left(-1, -\frac{\sqrt{2}}{2}\right)$

10. $\left(\tan \frac{1}{2}, \cot 1\right]$



Inverse Trigonometric Functions : RACE-06

Single Correct Question

1. $\tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \tan^{-1} \frac{1}{x^2 + 7x + 13} \dots$ upto n

terms.

(A) $\tan^{-1} (x + n) - \tan^{-1} x$

(B) $\tan^{-1}(x + n - 1) - \tan^{-1} x$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

MIT201

2. $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{2}{9} + \dots + \tan^{-1} \frac{2^{n-1}}{1 + 2^{2n-1}} + \dots$ upto infinite terms

(A) $\frac{\pi}{2}$

(B) $\frac{3\pi}{2}$

(C) $\frac{\pi}{4}$

(D) π

MIT202

3. $\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2} - 1}{\sqrt{6}} + \dots + \sin^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} + \dots$ upto infinite terms

(A) $\frac{\pi}{2}$

(B) $\frac{3\pi}{2}$

(C) $\frac{\pi}{4}$

(D) π

MIT203

4. $\cot^{-1} 7 + \cot^{-1} 13 + \cot^{-1} 21 + \cot^{-1} 31 + \dots$ to n terms.

(A) $\operatorname{arccot} \left[\frac{2n+5}{2n} \right]$

(B) $\operatorname{arccot} \left[\frac{2n+6}{n} \right]$

(C) $\operatorname{arccot} \left[\frac{2n+5}{n} \right]$

(D) $\frac{\pi}{4}$

MIT204

5. $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{1}{\sqrt{65}} + \sin^{-1} \frac{1}{\sqrt{325}} + \dots + \sin^{-1} \frac{1}{\sqrt{4n^4 + 1}} + \dots$ ∞ terms.

(A) $\frac{\pi}{2}$

(B) $\frac{3\pi}{2}$

(C) $\frac{\pi}{4}$

(D) π

MIT205

6. If $x \in (0, 1)$ and $f(x) = \sec \left\{ \tan^{-1} \left(\frac{\sin(\cos^{-1} x) + \cos(\sin^{-1} x)}{\cos(\cos^{-1} x) + \sin(\sin^{-1} x)} \right) \right\}$, then $\sum_{r=2}^{10} f\left(\frac{1}{r}\right)$ is

(A) 54

(B) 52

(C) 108

(D) 27

MIT206

7. Find the value of $3 \sum_{n=1}^{\infty} \left\{ \frac{1}{\pi} \sum_{k=1}^{\infty} \cot^{-1} \left(1 + 2 \sqrt{\sum_{r=1}^k r^3} \right) \right\}^n$

(A) 2

(B) 1

(C) 3

(D) $\frac{1}{2}$

MIT207

Matrix Match Type Question

8. Solve for x :

Column - I

(P) $\cos (2 \sin^{-1} x) = \frac{1}{3}$

(Q) $\cot^{-1} x + \tan^{-1} 3 = \frac{\pi}{2}$

(R) $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$

(S) $\sin^{-1} x + \sin^{-1} 2x = \frac{2\pi}{3}$

Column - II

(1) $\pm \frac{1}{\sqrt{2}}$

(2) $x = \frac{1}{2}$

(3) $\pm \frac{1}{\sqrt{3}}$

(4) $x = 3$

MIT208

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	A	C	A	C	C	A	B	$P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$